

November 15, 2024

Midterm

Duration: 90 minutes

Name:

Student No:

P1 [20 points] Construct a DFA for the languages(a) $\{w \mid w \in \{0, 1\}^* \text{ and } w \text{ has even number of 0s and contains at most two 1s.}\}$ (b) $\{w \mid w \in \{0, 1\}^* \text{ where } w \text{ is a binary number divisible by 5.}\}$ **P2 [20 points]** Construct an NFA for the languages below. Use as few states as you can. Fewest gets full score, every extra state halves your score.(a) $L = \{w \mid w \in \{a, b, c\}^*, |w| \geq 2, \text{ and the last two symbols of } w \text{ are the same.}\}$ (b) $L = \{w \mid w \in \{a, c, g, t\}^* \text{ where the symbols } a \text{ and } t \text{ alternate when the symbols } c \text{ and } g \text{ are ignored}\}$. For example $catgatgga, cggc, \varepsilon \in L$ but $cgatgacact \notin L$ (because $ataat$ has two consecutive as .)**P3 [15 points]** Prove that the language $\{0^k 1^m 0^n \mid \text{where } k > m > n > 0.\}$ is not regular.

P4 [30 points] Give the CFG and PDA for the following language: $\{a^k b^m c^n \mid k, m, n \geq 0, k \geq m + n\}$

CFG(15pts):

PDA(15pts):

P5 [15 points] Prove that the following language is not context-free:

$$L = \{0^m 1^n 0^k \mid k, m, n \geq 0, k = mn\}$$